EXPERIMENT 6

DATA VISUALIZATION WITH SEABORN

AIM:

To visualize and analyze the given dataset using various plots such as bar plots, count plots, histograms, matrix plots etc.. with the help of the Seaborn library, for better interpretation and understanding of trends and patterns.

PREREQUISITES & REQUIREMENTS:

1. Computer with Python Installed
2. A dataset for Practice
3. Jupytor Notebook
4. Knowledge on Pandas, Matplotlib & Seaborn Libraries

MATPLOTLIB:

- Seaborn is a Statistical library designed to interact with Pandas Dataframes.
- It is used to create common statistical plot types.
- It uses a simpler one-line syntax -sns.scatterplot(x='x_label',y='y_label',data=dataframe)
- Since, seaborn is built with matplotlib, so still we use plt method to create plots.
- Choosing the type of plot depends on the questions and relations we are working with.
- For more useful info about seaborn - https://seaborn.pydata.org
- While working with seaborn it should be imported - import seaborn as sns

Step 1: Import the required libraries & dataset

In [1]:
```
# Import the libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

In [2]:
```
# Load the dataset
_241UDS0092_df = pd.read_csv(r"C:\Users\hi\Downloads\monthly_sales_matplotlib_dataset.csv")
```

In [3]:
```
_241UDS0092_df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 4 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Month          12 non-null     str   
 1   Sales          12 non-null     int64 
 2   Profit         12 non-null     int64 
 3   Products_Sold  12 non-null     int64 
dtypes: int64(3), str(1)
memory usage: 516.0 bytes
```

In [4]:
```
_241UDS0092_df.head()
```

Out[4]:

	Month	Sales	Profit	Products_Sold
0	January	45000	8000	320
1	February	52000	9500	350
2	March	48000	8700	330
3	April	61000	12000	410
4	May	58000	11000	390

In [32]:
```
# Month wise data points in the dataframe
_241UDS0092_df.value_counts('Month')
```

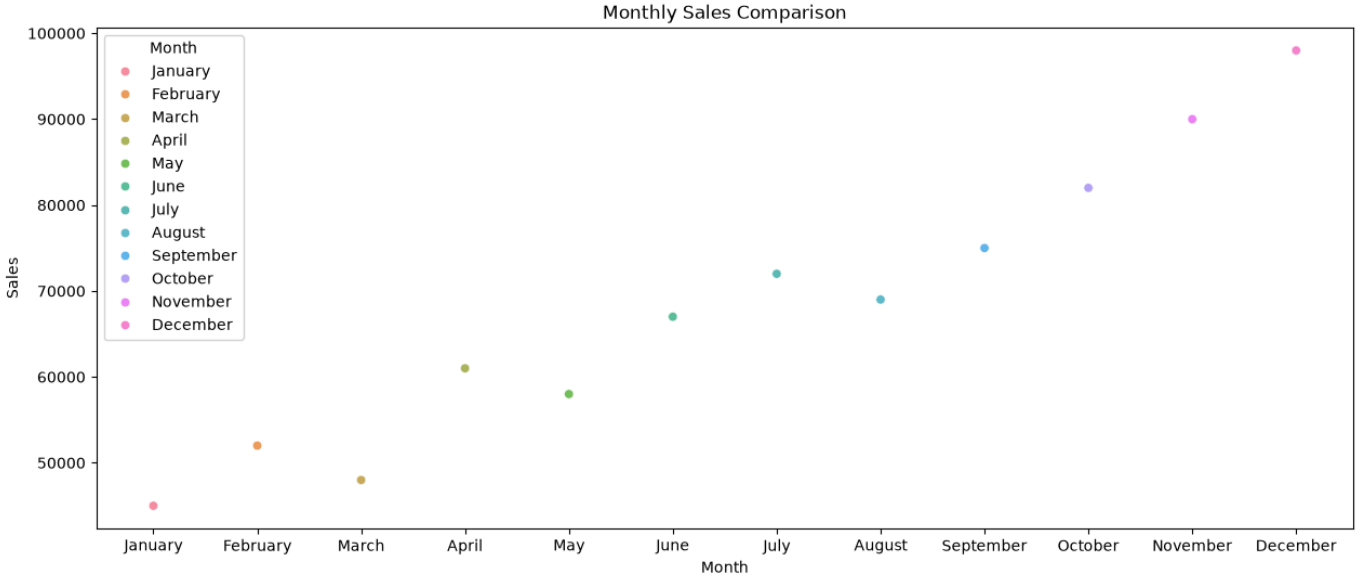
Out[32]:
```
Month
January      1
February     1
March        1
April        1
May          1
June         1
July         1
August       1
September    1
October      1
November     1
December     1
Name: count, dtype: int64
```

Step 2: Visualize Data Using Various Plots

Scatter Plot

Sales and Profit Comparison Over Months

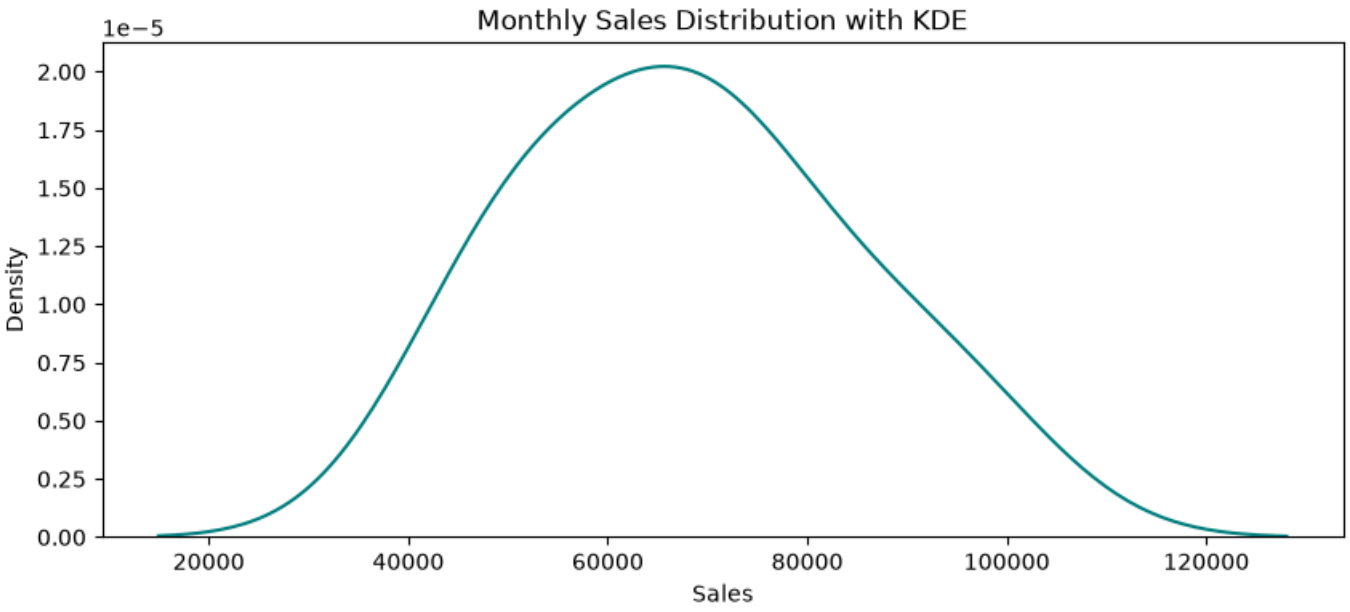
In [23]:
```
plt.figure(figsize=(15,6))
sns.scatterplot(x="Month", y="Sales", data=_241UDS0092_df, hue="Month", alpha=0.8)
plt.title("Monthly Sales Comparison")
plt.show()
```



Histograms & KDE

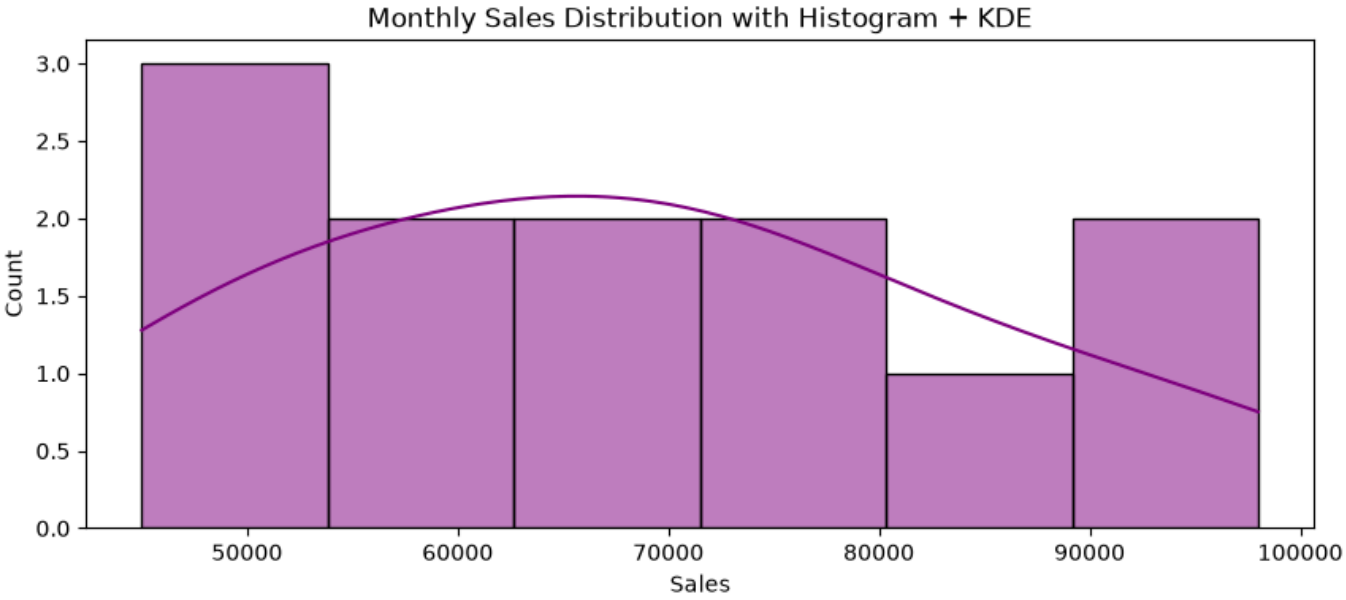
Sales Distribution with Histograms & KDE

In [24]:
```
plt.figure(figsize=(10,4))
sns.kdeplot(data=_241UDS0092_df, x='Sales', color='teal')
plt.title("Monthly Sales Distribution with KDE");
```



Histogram + KDE

In [36]:
```
plt.figure(figsize=(10,4))
sns.histplot(data=_241UDS0092_df, x='Sales', bins=6, kde=True, color='purple')
plt.title("Monthly Sales Distribution with Histogram + KDE");
```

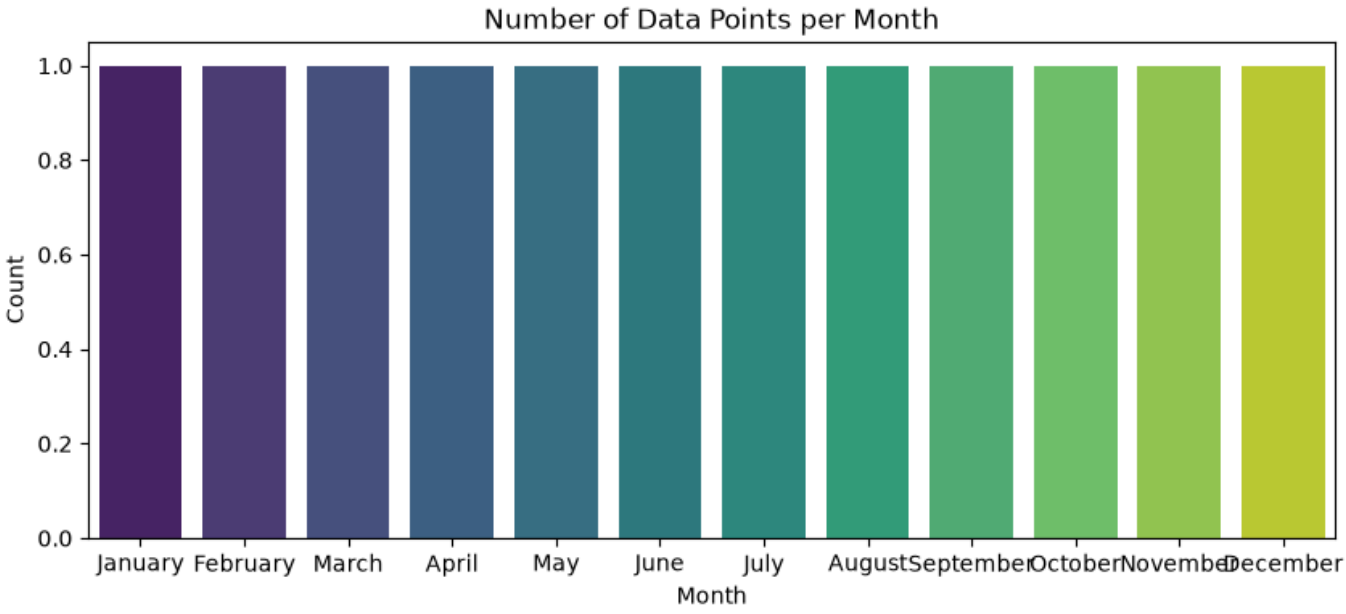


Count Plot

Number of Data Points for Each Month

In [37]:
```
plt.figure(figsize=(10,4))
sns.countplot(x="Month", data=_241UDS0092_df, hue="Month", palette="viridis")
plt.title("Number of Data Points per Month")
plt.xlabel("Month")
```

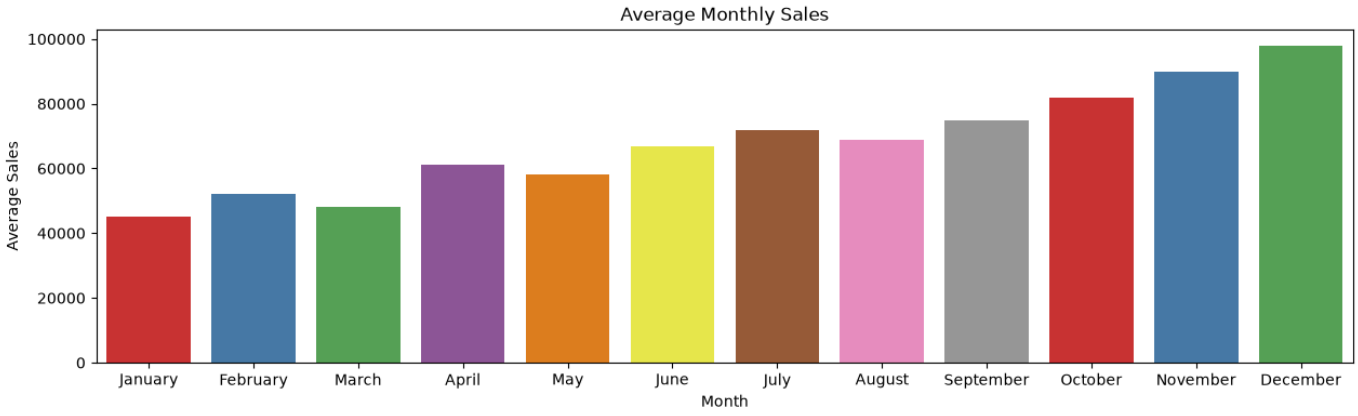
```
plt.ylabel("Count")
plt.show()
```



Barplot

Average Sales per Month

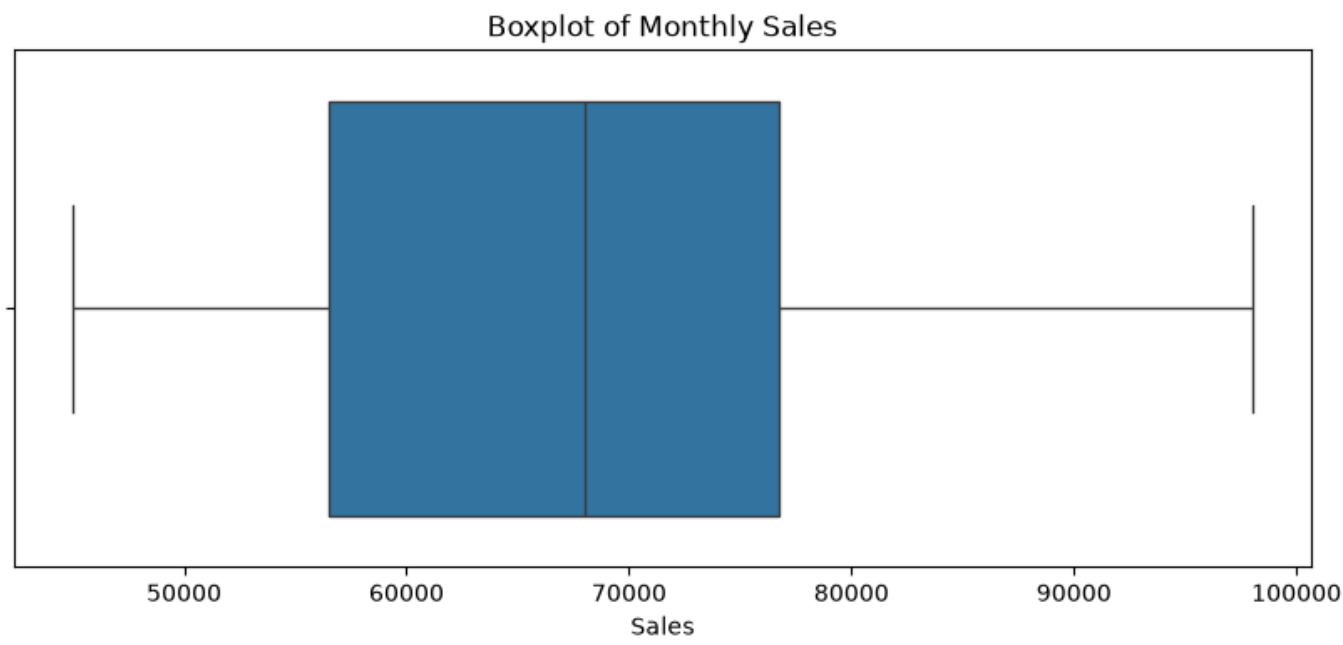
In [42]:
```
plt.figure(figsize=(15,4))
sns.barplot(x="Month", y="Sales", data=_241UDS0092_df, hue="Month", estimator=np.mean, palett
plt.title("Average Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Average Sales")
plt.show()
```



BoxPlot

Finding IQR's & Outliers of Sales Distribution

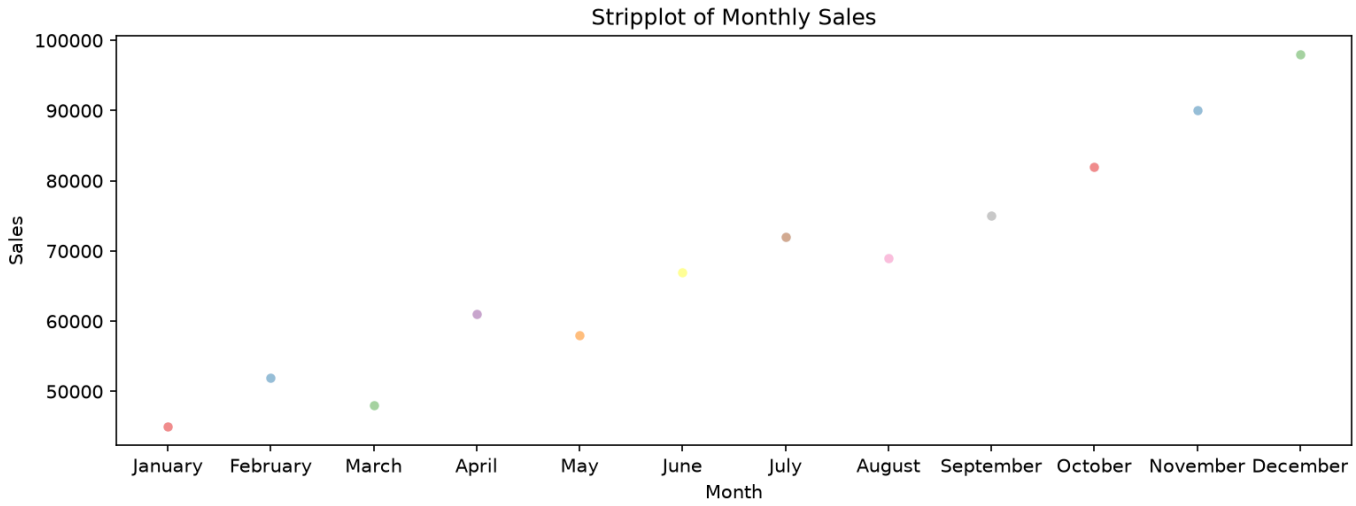
In [43]:
```
plt.figure(figsize=(10,4), dpi=100)
sns.boxplot(x="Sales", data=_241UDS0092_df)
plt.title("Boxplot of Monthly Sales")
plt.xlabel("Sales")
plt.show()
```



Swarm / Strip Plots

Finding Sales Distribution for Different Months

In [44]:
```
plt.figure(figsize=(12,4), dpi=150)
sns.stripplot(x="Month", y="Sales", data=_241UDS0092_df,hue="Month", palette="Set1", alpha=0.
# sns.swarmplot(x="Month", y="Sales", data=_241UDS0092_df, hue="Month", palette="Set1", alpha=
plt.title("Stripplot of Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

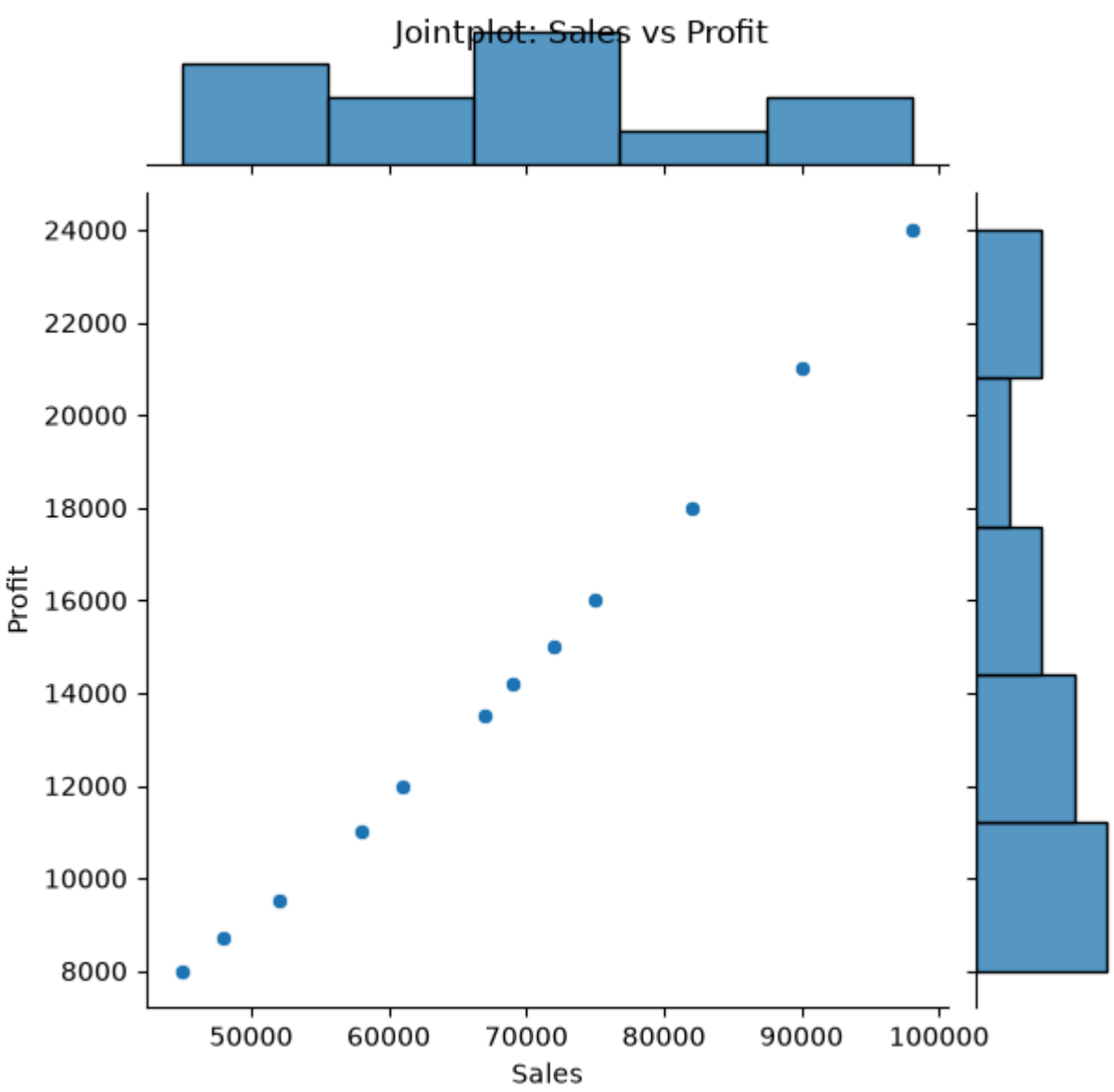


Joint Plots

Finding Distributions with Joint Plots

In [46]:
```
plt.figure(figsize=(18,6), dpi=150)
sns.jointplot(x="Sales", y="Profit", data=_241UDS0092_df)
plt.suptitle("Jointplot: Sales vs Profit")
plt.show()
```

```
<Figure size 2700x900 with 0 Axes>
```



RESULT

Various data visualization techniques using the Seaborn library were successfully implemented to analyze and compare monthly sales, profit, and products sold using different visualization plots.

In [ ]: